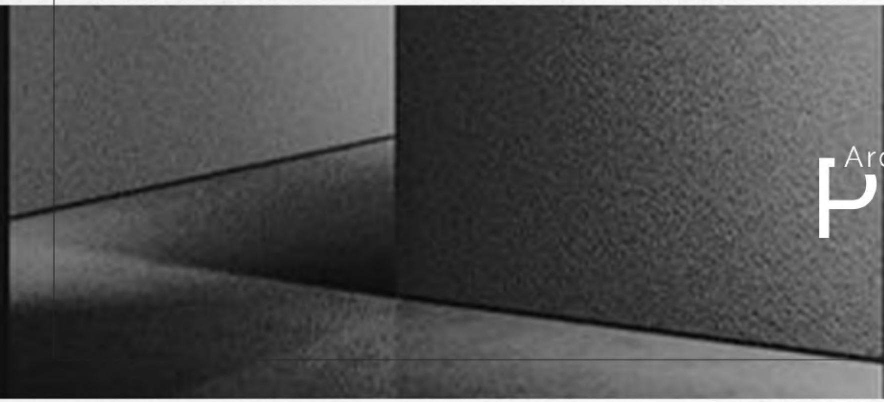


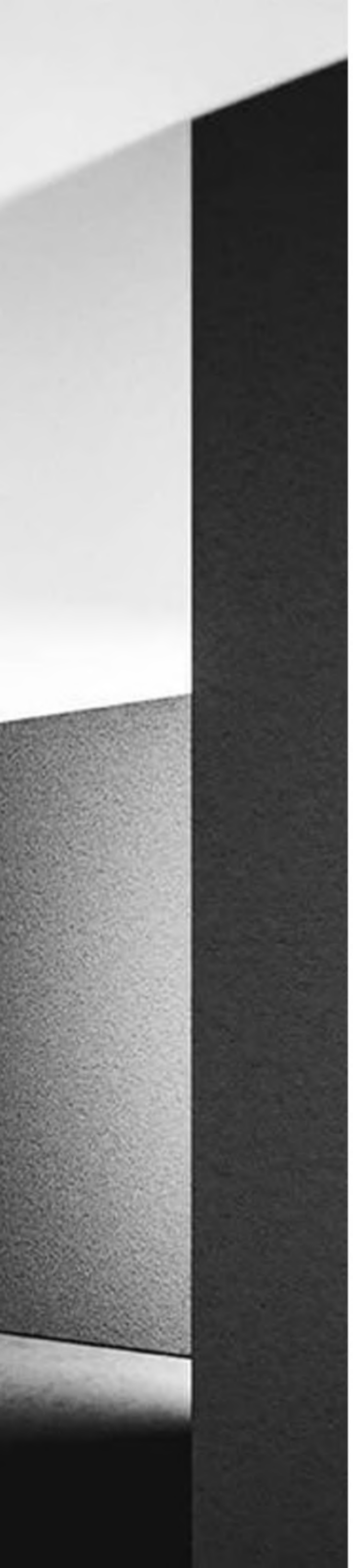
Architectural

PORTFOLIO

Maedeh (Kania) Kazemi

KANIA MAEDEH
M.Sc ITBE KAZEMI
12th June 1999

"This portfolio reflects my early journey in computational design and digital fabrication as the beginning of my path toward digitalization and automation, with robotics and simulation at its heart."



Content

01

Technology Hub

Final Studio Design

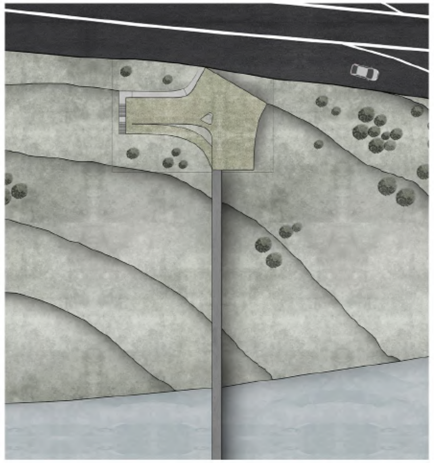
Academic Project | University of Tehran | 2022
Supervisor: Dr. Saeed Haghir
Individual Project

The idea of designing a multi-purpose complex, emphasizing collective spaces and activities for all age groups, inclines design direction towards flexible and creative spaces. Defining a dynamic space in interaction with the city, with a future-oriented view and providing a platform for the growth and emergence of creativity, along with recreation and relaxation, is an issue that is of great importance to address. Examining a strategic issue from the point of view of the city, site, and users, along with solving the problem, has led to the creation of practical space for the future and will bring a new chapter of the issues in the design of buildings. Linking contacts from different spectrums brings a new kind of reading for studying functional spaces. In a mixed space, the knowledge of coexistence is far more critical than selective choice, and in the modern world, it emphasizes the importance of users twice. The perspective of the combination of users and spaces in a boundless collection, on the one hand, will define architecture as functionalist, and on the other hand, it will define dynamism and mobility in the environment.

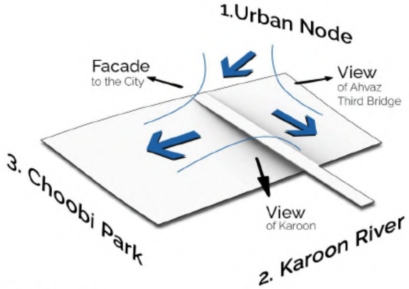
Design Process

The design process is defined in 14 diagrammatic sequences. The form-finding process is entirely based on the main concerns and goals of the design and is the result of considering the natural and human forces around the site. In this way, the design process starts with defining the spatial forces and sequences of the site. These forces - urban node, Karoon River, and wooden park - have led to the initial placement of mass and void in the site; each placement has its own strategy. The void was directly influenced by the first force that formed the main entrance of the project and the direct access to the Tabiat bridge. The first mass is the result of the first and second forces, and the second mass is the result of the first and third forces. The placement of masses and void defines three fronts. These three fronts provide a view of the Karoon River, a view of the Third Bridge of Ahvaz, and a view of the project towards the city. The placement of the masses is in such a way that the maximum distance from the edge of the river is formed to respect the privacy of the river. Moreover, the levels of the edges of the masses are redefined to be aligned with the level of the site and Tabiat bridge, and then the bridge zone is expanded to form the first public plaza on the ground floor. The next steps are followed logically as well to complete the design process.

Location: Ahvaz, Khozestan, Iran



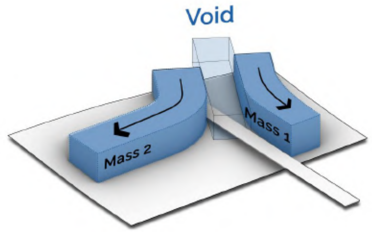
Site Plan



01

Analysis of Site Forces and their Results

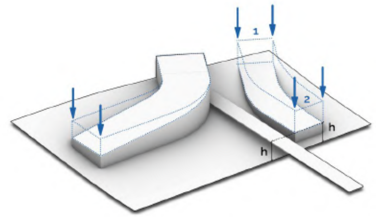
- Force #1: Urban Node → Bringing city to the site
- Force #2: Karoon River → View of the Karoon river
- Force #3: Choobi Park → View of and access to the park



02

First Location of Mass and Void

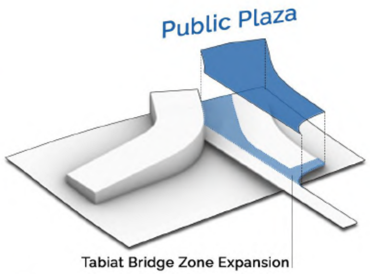
- Void: Force #1 → Entrance and access to the Bridge
- Mass #1: Force #1 + #2 → View and access to the river
- Mass #2: Force #1 + #3 → View and Access to the Park



03

Redefining Masses' Edges

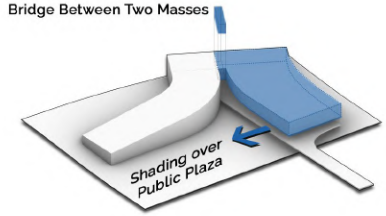
- Mass #1 → Lowering the first edge to the city level
  → Lowering the second edge to the level of the bridge
- Mass #2 → Lowering the edge closer to the park level



04

Bridge Zone Expansion

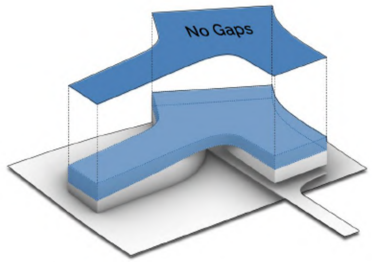
- Plaza #1 → Creating a public plaza at the ground level by adding a volume between the bridge and mass #1



05

Creation of First Floor

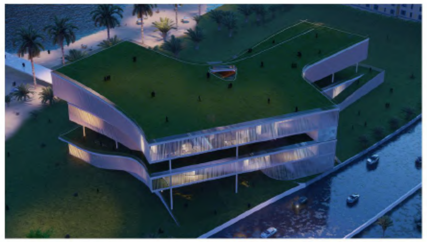
- Adding a new volume using the lines forming the first mass
- Adding a volume to connect the new volume and mass #2

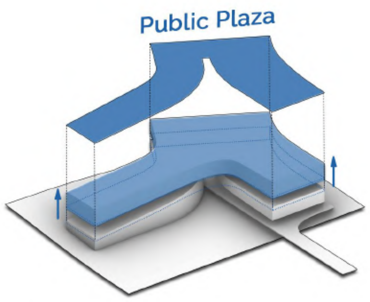


06

Creation of Second Floor

- Creating a new volume using the perimeter lines of mass #1 and #2 without any empty space to achieve the maximum area for the main use

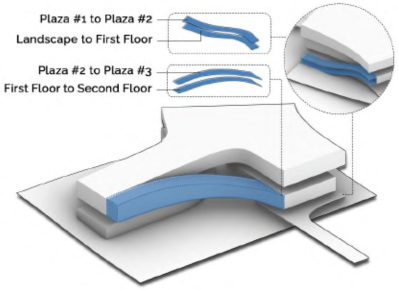




07

Creation of Plaza #2

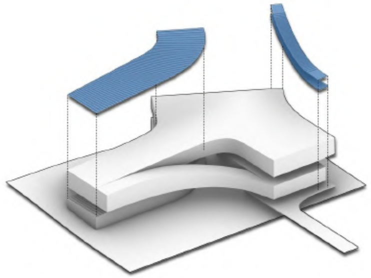
- Changing the level of the second floor upwards to create Plaza #2 between the first and second floors
- Creating a semi-open space favorable in terms of climatic conditions by shading the volume of the second floor



08

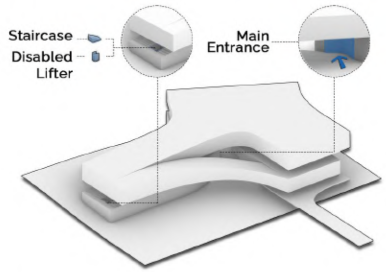
Connection of Spaces and Plazas

- Connection between closed spaces with each other
- Connection between the plazas with each other



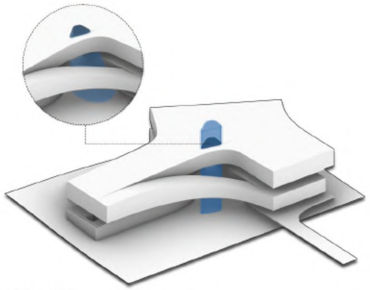
09

Defining Outdoor Connecting Stairs



10

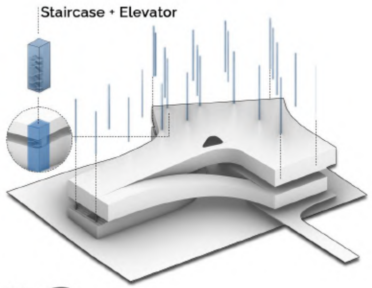
Creation of Entrances



11

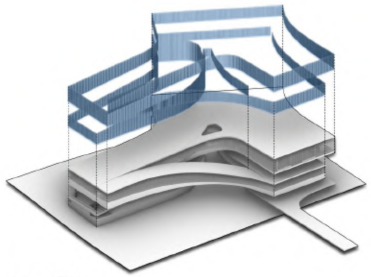
Creation of Atrium

- Illumination to Plaza #1 and the Bridge



12

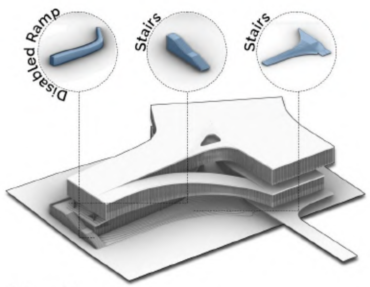
Structure and Vertical Circulation



13

Defining External Walls and Louvers

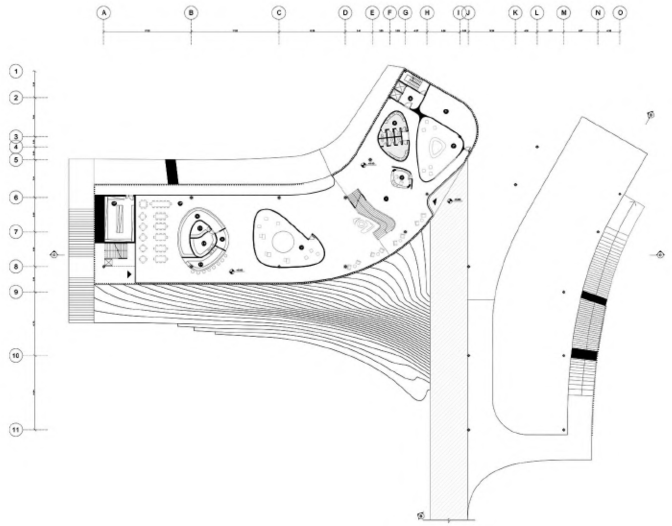
- Using glass in all windows to achieve maximum view
- Using louvers to control light and heat in response to climatic restrictions



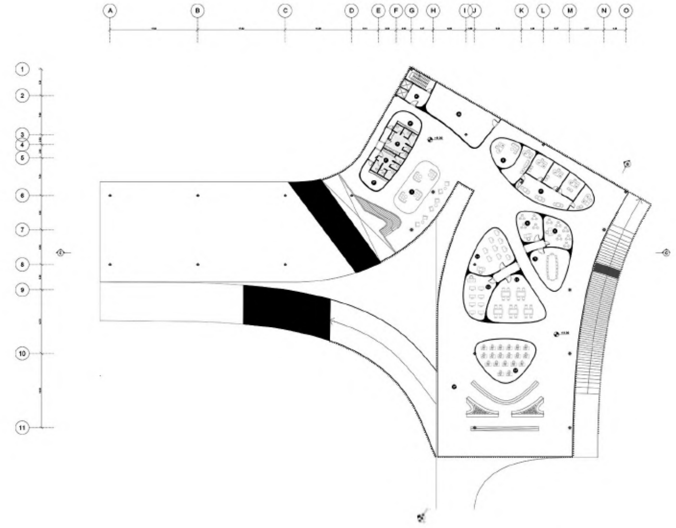
14

Organizing the Landscape

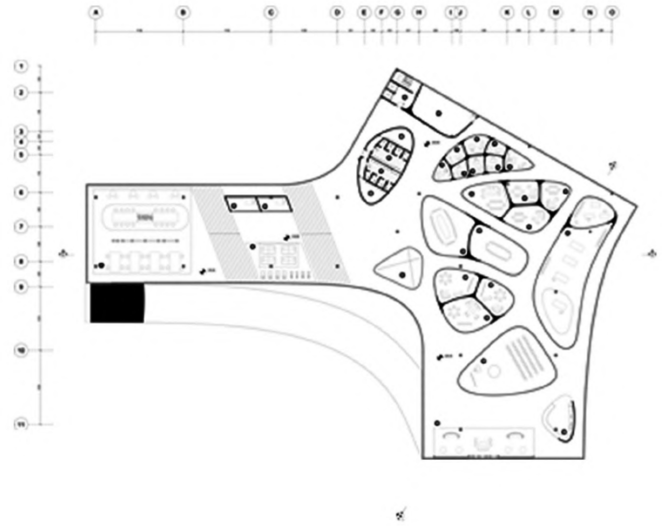
- Using ramps and stairs as connecting elements between the site and the building



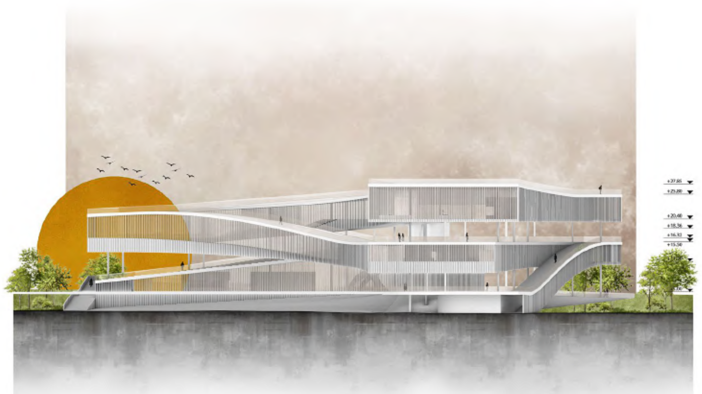
Ground Floor Plan



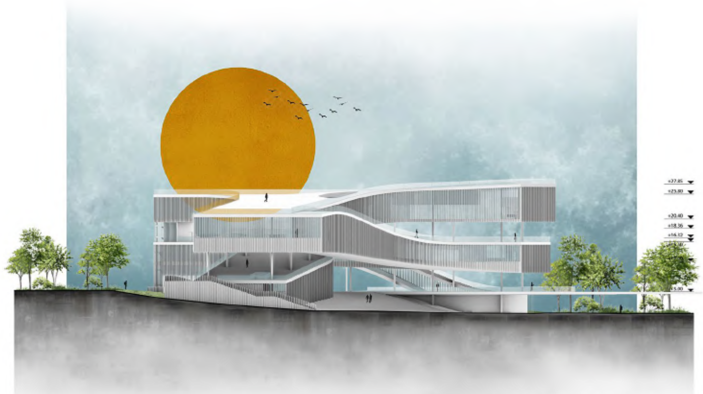
First Floor Plan



Second Floor Plan



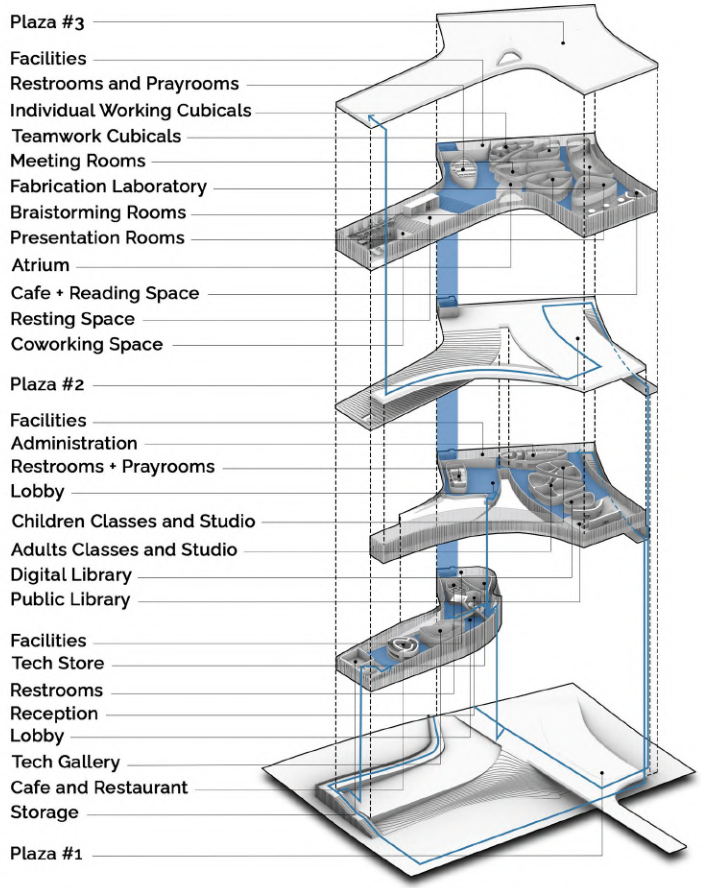
East Elevation



South Elevation

Section A-A

Section B-B



Indoor Perspectives

• Lobby | Ground Floor

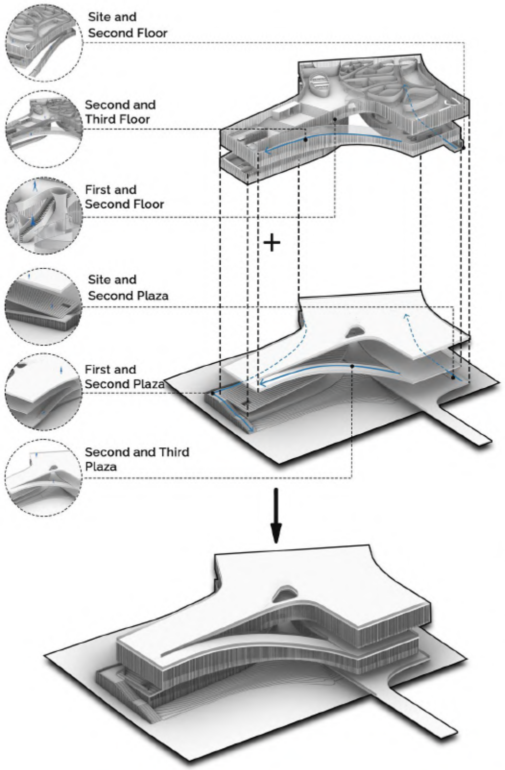
• Lobby | Ground Floor

• Public Library | First Floor

• Cowork Space | Second Floor

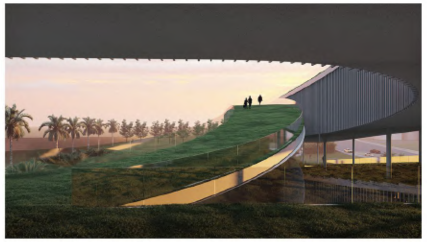
• Resting Cubicals | Second Floor

• Reading Space-Cafe | Second Floor



Vertical Circulation

One of the important points in defining two movement scenarios for this complex is to control access to floors and plazas so that they do not interfere with each other. The first movement scenario reaches the open and semi-open spaces and moves the users between the plazas in a continuous path. This movement is without any conflict with closed spaces and directs the users to the plazas on the roof of the floors. With the same strategy, circulation among the closed spaces happens without engaging with the plazas and open spaces. Through internal paths, the specific users of each floor follow the consistent and regular path of circulation among the spaces and use internal uses.




36
B
C
32
A
B
C

02

Floated Plates

Computer Aided Design and Fabrication (CADF98)

Academic Project | University of Tehran | 2019
Instructor: Dr. Ramtin HaghNazar
Team Project | Contribution: Form Finding, Coding, Execution

CADF98 was a 2-credit course for undergraduate students at the University of Tehran in 2020. This course is defined to introduce integrated computational design and fabrication to students. They learned about the basic logic of the computational design and Grasshopper software. Then, students try to make a preliminary sketch of an installation. In the final sector, students finalized the design and developed a Grasshopper code to generate a model and shop drawing. The structure is made of a cubic frame which is 2.4×2.4×2.4 m^3, and a number of suspended panels fixed with cables. All the parts were cut from 9mm thick plywood sheets by a CNC milling machine and assembled by students in the court of the Faculty of Fine Arts.

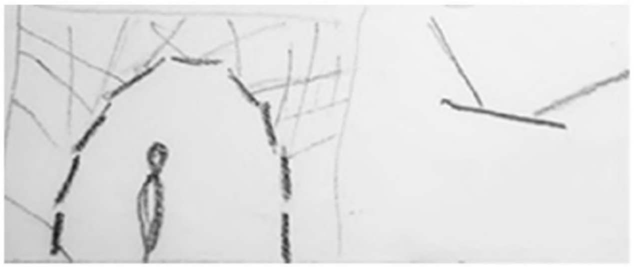
Introduction

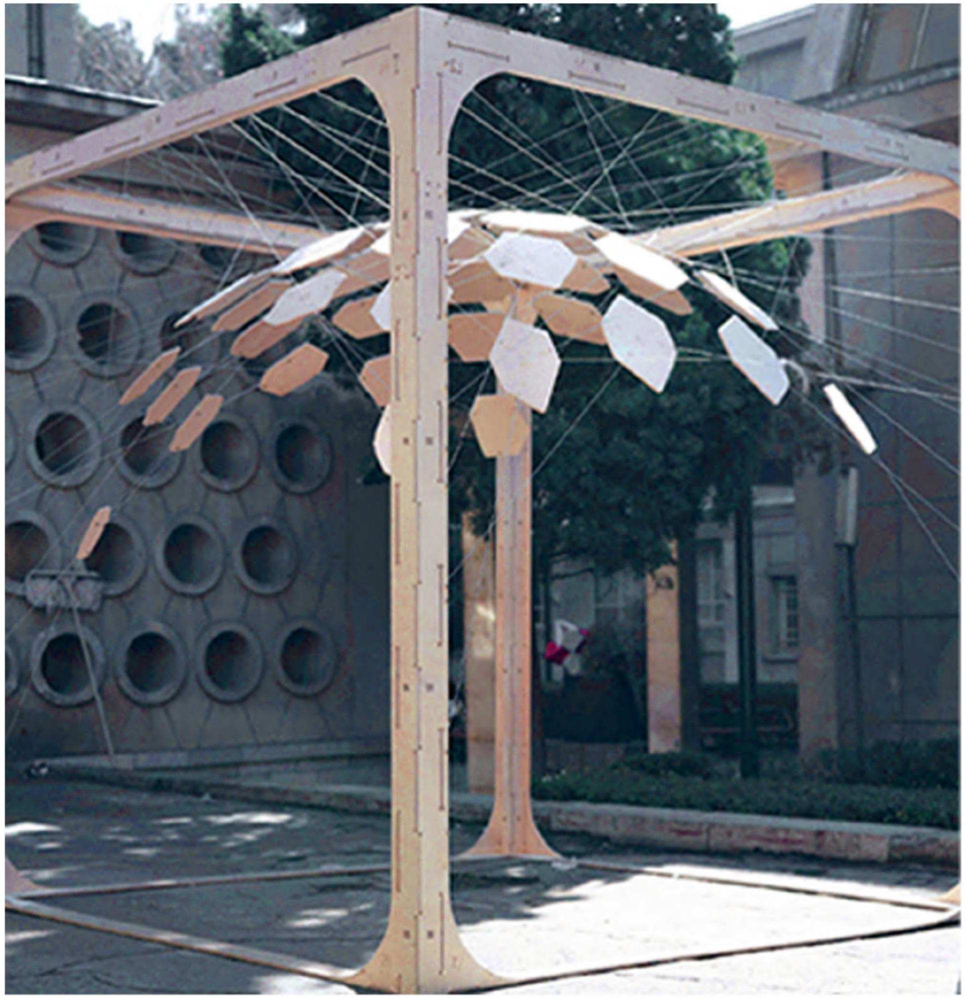
CADF98 was a 2-credit course for undergraduate students at the University of Tehran in 2020. The purpose of this course was for students to experience integrated digital design and to get acquainted with digital fabrication contrivances and facilities. Our ultimate goal was to produce an installation by students in the court of the Faculty of Fine Arts. This installation was constructed from plywood sheets, and it uses a technique of suspended structures. All parts of this installation were digitally designed and later manufactured. This course was attended by 36 students and three teachers.

Process

The first phase of this course consisted of two parts: Training and experience. In the training section, students were introduced to the grasshopper software as a digital design and fabrication tool to realize their potential in design. They tried to build a simple structure. In the next step, students experienced integrated computational design and fabrication. They tried to design an installation and developed a grasshopper code to generate a model and shop drawing.

In the final phase, one of the students' models was elected for the final real-scale model. This proposal contained an installation in the court of the Faculty of Fine Arts. Here, students tried to find the best detail for the joints considering restrictions; the material was restricted to 9mm thick plywood, and the technique was limited to a CNC milling machine.





Main Frame

Generating the Structure

Pattern Drawing

Placing an Element

Suspended Panels with Cables

Final Product

After creating the model as illustrated, the team tried to make a G-code for the CNC milling machine to cut the elements. The structure is made of a cubic frame which is 2.4×2.4×2.4 m^3, and a number of suspended panels attached to it by cables. All the parts, cut from 9mm thick plywood sheets by a CNC milling machine, were assembled by students in the court of the Faculty of Fine Arts.

The first phase of the construction included the preparation and attachment of the parts of the cubic frame, which is 2.4×2.4×2.4 m3. For this purpose, the structural parts, which were prepared by the students to be connected to each other after they were cut with a CNC milling machine, were installed in the court of the Faculty of Fine Arts.

The exact location of each element is determined by numbering, and the length of the cables and the exact location of their closure is determined by the Grasshopper software. As a result, the cables were cut to the specified length according to the shop drawing, and after securing the frame, they kept the elements in the right place.

03

A. Covid-16

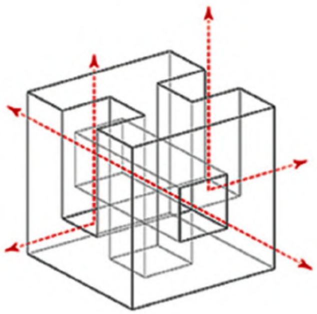
Minimal Surfaces | Product Design and Fabrication

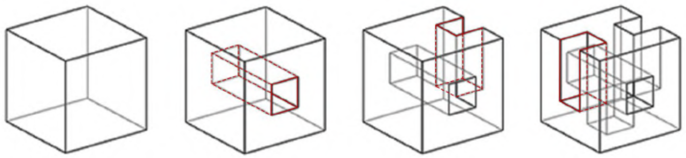
Professional Project | Marizad Architects | 2021-2022
Principal Architect: Mohsen Marizad
Team Project | Contribution: Research, Design, Fabrication, Presentation

Topological Surfaces facilitate connection and interaction due to their intrinsic properties of self-intersection and continuous deformation. Their morphology depends not on the shape of boundary extents but rather on the way they connect together. Therefore, the general geometric makeup is preserved even with continuous shape morphing. Capitalizing on these characteristics underlines the computational reasoning behind their use in design: topological surfaces have the potential to move between scales, seamlessly connecting interiors with exteriors and form with space. Inspired by these properties, this collection of objects uses the geometric framework as a design agent to explore connection and multiscalar performance in design. Topological Surfaces are created through 3-dimensional digital simulations of topological surfaces and are developed through a framework of generative processes based on functionality and specific performance factors. Through intricate and continuous morphological reformation and interconnection, the pieces suggest interaction and spatial connection. They are an instrument to explore the connection with others and the surrounding environment.

Creation of the Central Hub

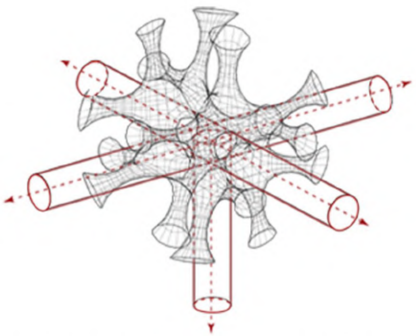
The core is created from a cube subtracted in its three sides, with three smaller cubes.

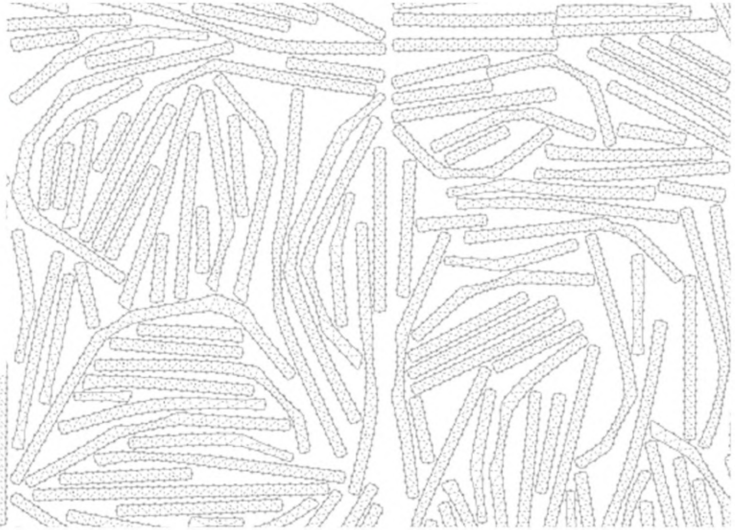




Central Voids

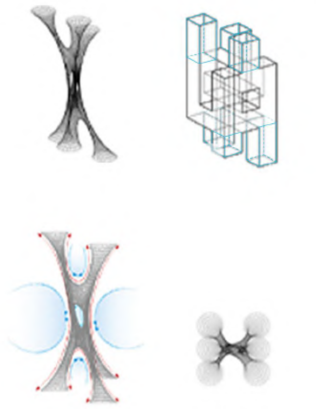
The central hub has three orthogonal voids in the X, Y, and Z axes, extended from one side to the other. As a consequence, the arms surround an empty core.





01

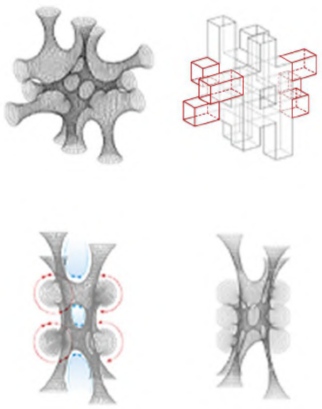
Connection of one hub and legs:
Hubs: 1
Primary Surfaces: 18
Secondary Surfaces: 24
Legs: 6



Front View   Top View

02

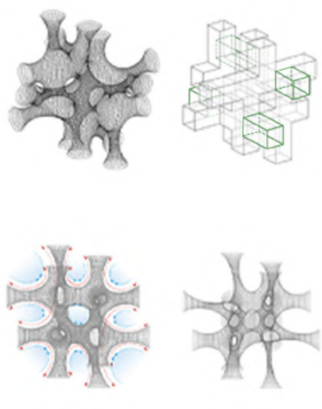
Connection of one hub and legs:
Hubs: 1
Primary Surfaces: 42
Secondary Surfaces: 24
Legs: 12



Front View   Top View

03

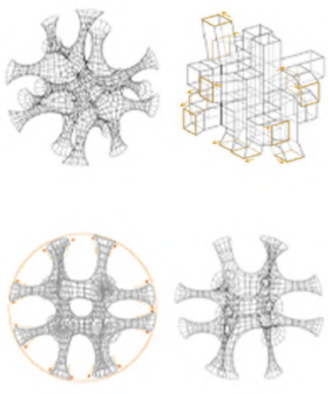
Connection of one hub and legs:
Hubs: 1
Primary Surfaces: 66
Secondary Surfaces: 8
Legs: 16



Front View   Top View

04

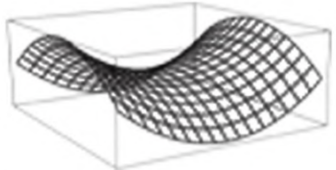
Connection of one hub and legs:
Hubs: 1
Final Surfaces: 72
Final Legs: 16

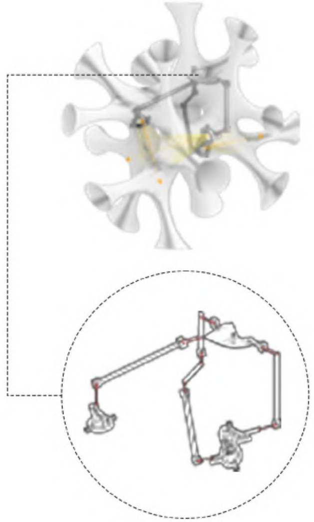


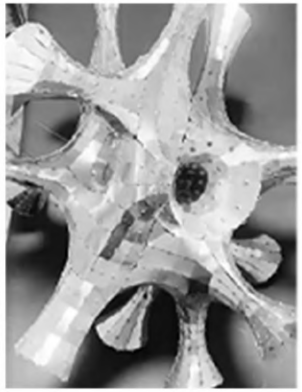
Front View   Top View

Arms Structure

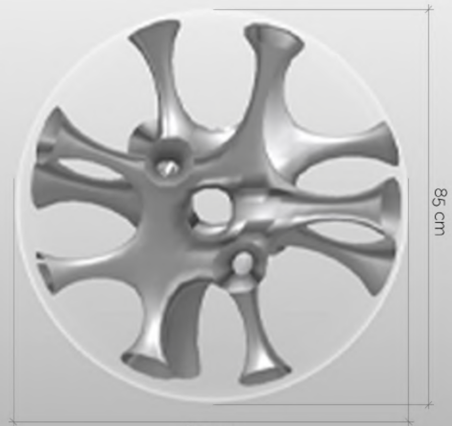
Covid-16 is designed with three light zones. Light bulbs of each zone are hung from a structure having three primary arms. These arms are connected to a hyperbolic (saddle-shaped) bed, and the whole lighting puts its weight on the bed.







Dimensions:	Encircled in a sphere of 42.5 cm radius
Material:	Aluminium
Color:	Silver
Light Source:	3*15W daylight bulb Non dimmerabile
Energy Efficiency:	A+



03

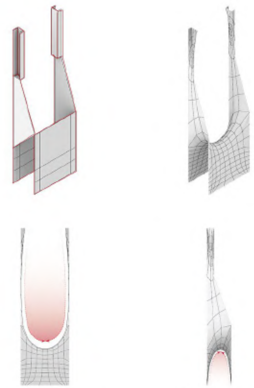
B. Inter-Void Tower

Minimal Surfaces | Tower Concept Design

Professional Project | Marizad Architects | 2022
Principal Architect: Mohsen Marizad
Team Project | Contribution: Research, Design, Fabrication, Presentation

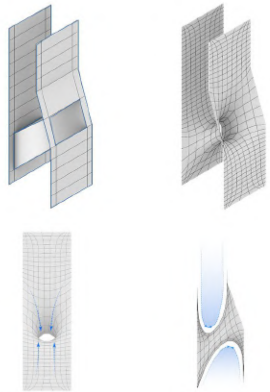
Topological Surfaces operate at the intersection of technology, structure, materiality, and a scalar spectrum from the intimate to the territorial. The collection represents an approach to design that tightly integrates form and space, simulation, and digital fabrication to create complex, novel pieces. These prototypes are created in 3-dimensional software through generative design processes based on functionality and performance. They are produced with digitally fabricated planar plates, and the tongue and groove joinery technique was used to make the final model.

01 Inner Shell



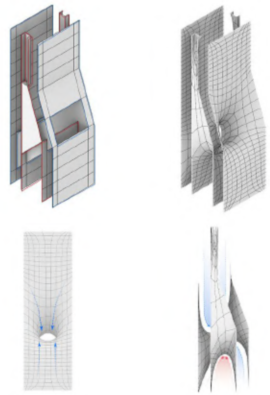
Front View Side View

02 Outer Shell



Front View Side View

03 Shells Intertwine



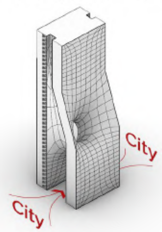
Front View Side View

Horizontal Void

A horizontal void splits the tower at levels close to the ground. Due to this horizontal void, the ground levels of the tower are assigned to the city and its residents, and it pulls the city into the tower. According to the urban design, this horizontal void can be an urban plaza or a street and its sidewalks, which continue under the tower. In any case, this horizontal void will provide the city and its citizens with a new experience.

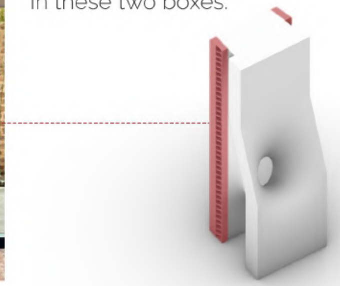
1971

2022



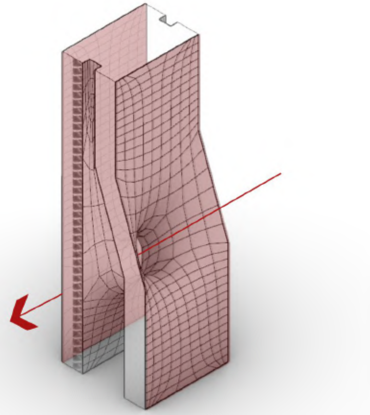
Vertical Circulation

For Easy access to top floors two boxes of vertical circulation are provided in two sides of the tower. Stairs and lifts are placed in these two boxes.



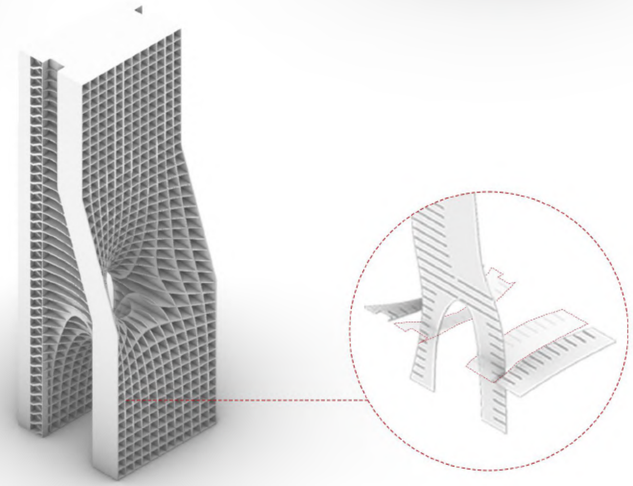
Inter-Void

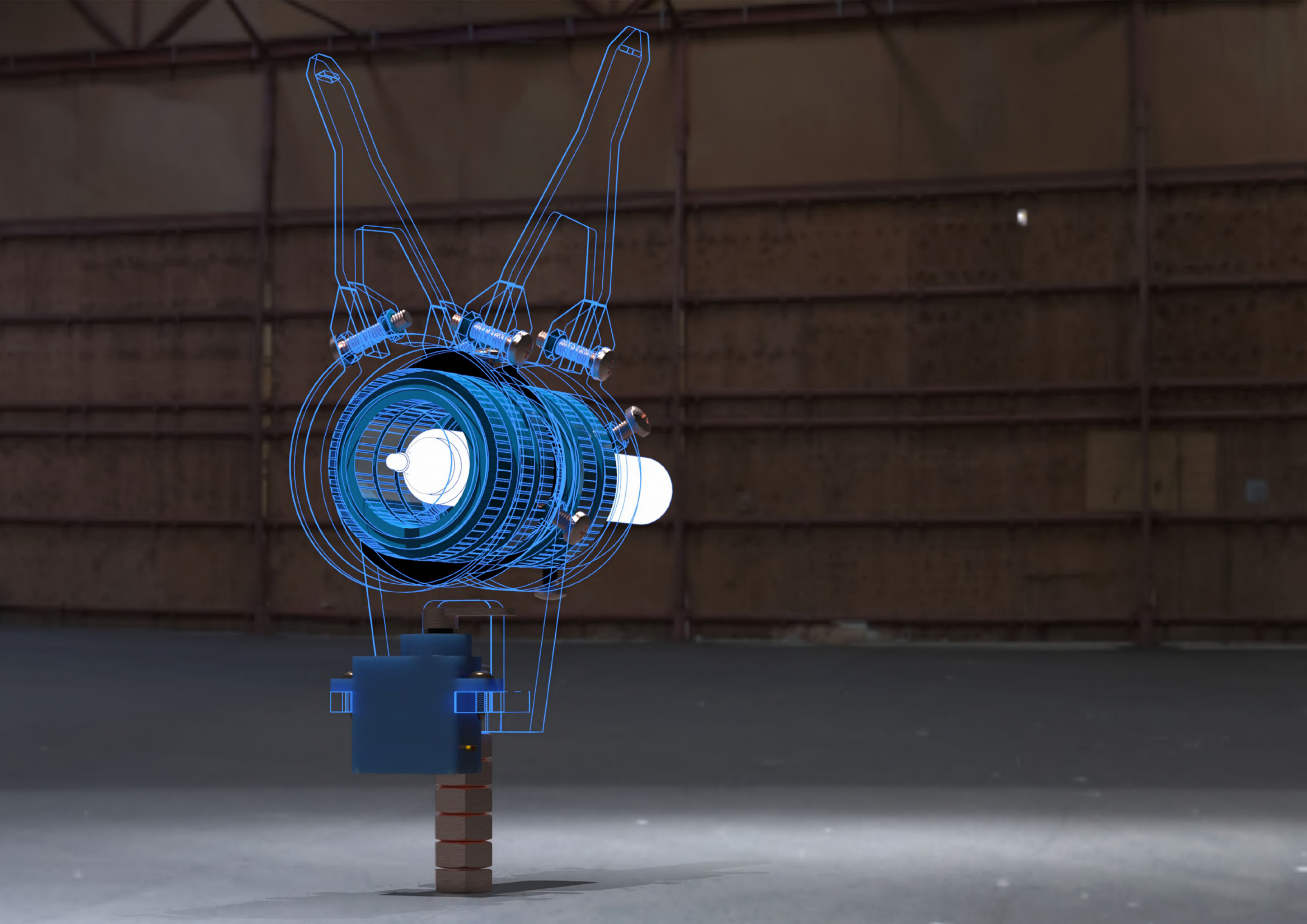
The middle void that diagonally connects one side of the tower's facade to the other introduces a new definition of the facade. In this definition, the facade is a three-dimensional shell, which is more dynamic than two-dimensional facades. In addition, this void forms a flow of light from one side of the tower to the other.



Fabrication Technique

The tongue and groove joinery technique was used to make the model. In this method, the mesh is divided horizontally and vertically into separate plates of 3mm thick plywood. Grooves are defined on the surface of the plates so that these plates are woven into each other and finally form the overall model of the tower.





04

Polar Plotter

Responsive Architecture

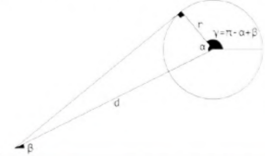
Academic Project | Held by Shahid Beheshti University | 2022
Instructor: Dr. Mahdiyar Esmailbeigi
Team Project | Contribution: Coding, Execution, Documentation

A polar plotter, also known as a polar graph or Kritzler, is a plotter that uses bipolar coordinates to produce vector drawings using a pen suspended from strings connected to two pulleys at the top of the plotting surface. This gives it two degrees of freedom and allows it to scale to relatively large drawings simply by moving the motors further apart and using longer strings. Some polar plotters will integrate a raising mechanism for the pen, which allows lines to be broken while drawing.

This project is a polar plotter that uses mathematical logic to find bipolar coordinates and create images using a pen suspended from strings connected to two pulleys at the top of the plotting surface. For plotting, it takes input from the output of the Javid Plugin in Grasshopper and can change its drawing behavior accordingly.

Mathematical Logic of the Pulley's Rotation

In order to find out how many degrees each pulley should be rotated by the stepper motor, a tangent from the pen point to the center of the pulley's circle is drawn.
Two angles, **α** and **β** are created by connecting the center of the pulley's circle to the pen point.
To find the angle of rotation, i.e.,**γ**; First, get the pen distance to the center of the circle, i.e., d.
r is the radius of the pulley and creates the **α** angle with d.

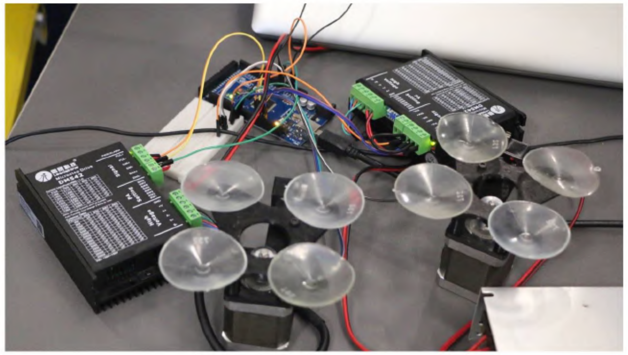


α is equal to: $\alpha = \arccos r/d$

β is equal to the x vector of the d vector.

Therefore, γ is equal to: $\gamma = \pi - \arccos r/d + \beta$

```
private void RunScript(Circle circle, Point3d pt, ref object A)
{
  var line = new Line(pt, circle.Center);
  var alpha = Math.Acos(circle.Radius / line.Length);
  var beta = Vector3d.VectorAngle(line.Direction, Vector3d.XAxis);
  A = circle.PointAt(Math.PI - alpha + beta);
}
```



Implementation Steps and Details

After reviewing different case studies, projects, and their components, the following software, components, and details were designed or used for the project:

- Softwares

1. Arduino IDE
2. JAVID (A plugin for Rhino/Grasshopper)

- Optimization

During the process of assembling the plotter, to optimize the trigonometric calculations (accuracy), the most optimal method of installing the pen holder to the cable and optimized weight was used.

- End-stop Sensor

This switch is used to find the zero coordinates of the foreground screen based on the distance of the steppers and their height.

- Glass-Background

Using glass as a background is a unique concept and makes the project more noticeable and interesting for the audience. By using the shop window glass and mirroring the image in the Grasshopper, the drawn image will be displayed in the original direction for passers-by.

- Gondola

The pen holder was designed and printed in its current format after several changes and tests. After calculating the dimensions, all the non-metallic parts were 3d printed and joined with different components, such as the ball bearings or screws.

- JAVID

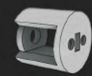
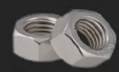
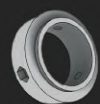
For Grasshopper in Rhino: JAVID is an open-source graphic design and image-processing Grasshopper plugin developed in the Coding for architects workshop at IAC. It can represent Images using different graphical techniques, such as ASCII Art, String Art, and many others. This project uses this plugin to illustrate the line art pictures to ease the drawing process. Javid was designed and written by Dr. Mehdiar Esmailbeigi and the students of the "Plugin Development with C# Programming Language course.

Hardware

The following parts were used to assemble this project:

Number	Name of the piece	Application	Image
01	Arduino Uno	The main part	
02	Jumper Wires	Connects the Arduino board to the driver	
03	Stepper Motor	Moves in gear based on the location of the pen	
04	Engine Servo	Adjusts the distance between the tip of the pen and the surface	
05	End-Stop Switch	Finds the zero coordinates of the desired plane	
06	Stepper Motor Driver/Switching (+24V – 10A)	Converts pulse signals from the controller into motor motion to achieve precise positioning	
07	Voeding Driver Adapter- Practical Power Supply Driver	Supplies electric power	

Industrial Parts:

Number	Application	Image	Number	Application	Image
08	Ball bearing eases circulation of pen holder parts		16	Holds the stepper motor	
09	The strap makes a connection between the pen holder and the stepper motor holders to transfer and adjust the movement		17	Prevents the dislocation of the belt by the weight of nuts	
10	Used as weights		18	Connects part #20 to the main central piece	
11	Holds the servo motor		19	Joins belts to the gondola	
12	Holds the nuts that keep the gondola stable		20	Holds the end-stop switch	
13	Main central piece		21	Keeps the end of the belt and part #20 together	
14	Creates a smoother connection between part #17 and part #19		22	Balances both sides of the plotter	
15	Fixes the pen using screws		23	Adjusts the size for holding the pen	
			24	Sticks the stepper motor holders to the glass	

05

Form-Finding Explorations in CLT Construction

Workshop | Authorized by Augsburg University of Applied Sciences (Digital Timber Construction)
Held by **Acadia** and Weitzman Architecture, University of Pennsylvania | 2022
Instructor: Amin Adelzadeh, Hamed Karimian A.
Team Project | Contribution: Form-Finding

The application of Cross-Laminated Timber (CLT) in the building industry has a major role in shaping the recent renaissance in timber construction; however, despite the benefits, CLT fabrication produces a huge amount of offcuts that, despite the high quality, are too small for the timber building applications. For the first time, these CLT offcuts were turned into a segmented shell demonstrator, "Recycleshell," by using algorithmically-generated form-fit connectors and screws. Inspired by such a research philosophy, this workshop aimed to provide fundamental knowledge and essential computational skills for the design-to-assembly of lightweight segmented wood-only shell structures made of CLT plates. Participants gained knowledge of innovative CLT construction techniques, material systems, joinery, rapid and precise assembly, and improved computational skills in developing modeling workflows consisting of form-finding, discretization, rationalization, joint generation, and fabrication data production.

Modeling Workflow

Form-finding

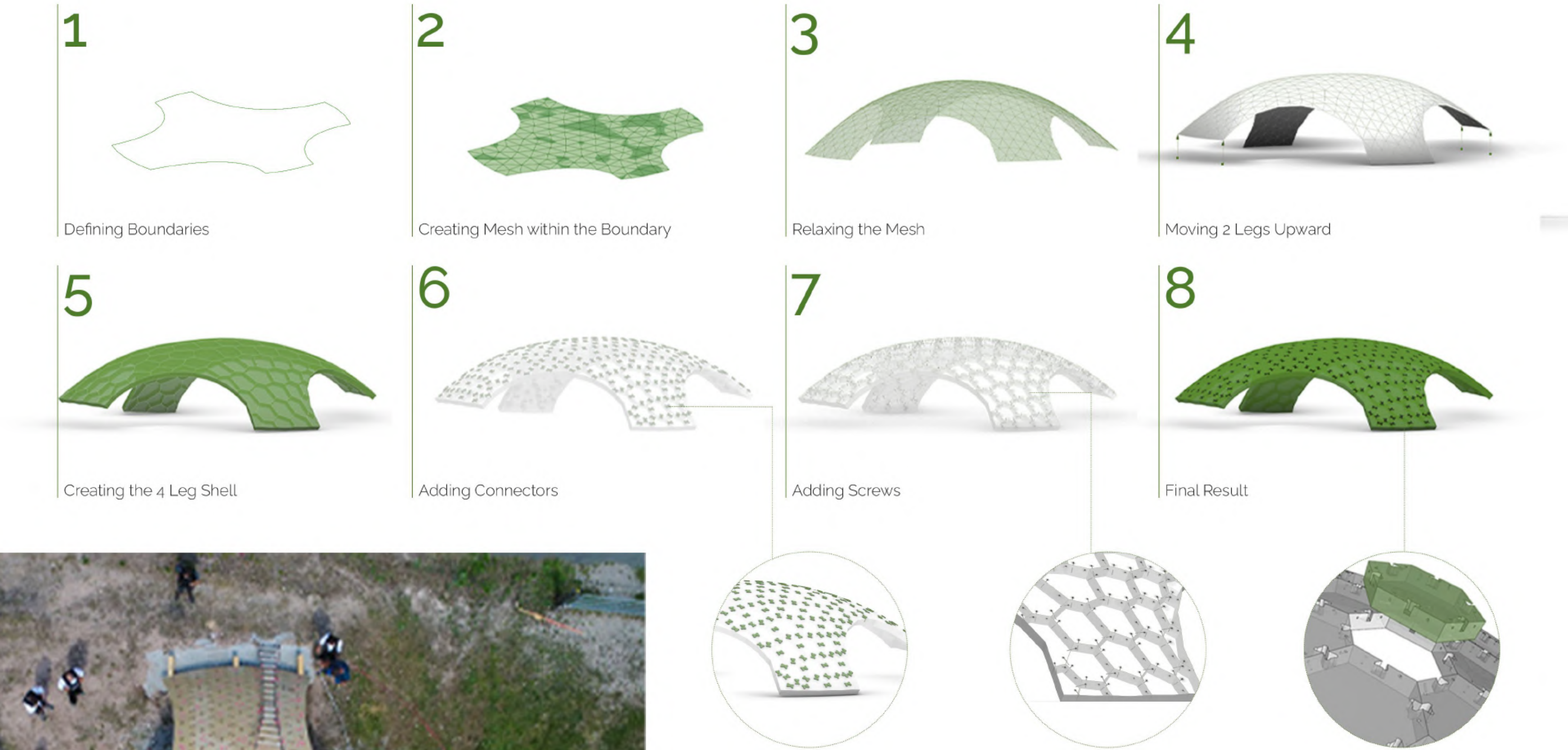
Rationalization

Plate Generation

Joint Application

Fabrication Data

Doors and window cutouts from large CLT blanks provide high-quality but small-sized sheet stock, which is difficult to design for and build using conventional methods. Using the predictable geometry of these offcuts as a design generator, however, can provide novel architectonic expression while further reducing the waste of valuable slow-growing mass timber material. In this exploration, the "Recycleshell" design-to-fabrication workflow is developed further to align the panelisation strategy with the expected size, structural performance, and material characteristics of typical CLT offcuts. The result is CLT shell segments scaled and proportioned to door and window cutouts, with the timber lamella and grain oriented for optimum structural performance.


1
Defining Boundaries
2
Creating Mesh within the Boundary
3
Relaxing the Mesh
4
Moving 2 Legs Upward
5
Creating the 4 Leg Shell
6
Adding Connectors
7
Adding Screws
8
Final Result

06

Active Bending Shells

Freeform Space Structure (FSS #5)

Workshop | Held by Digital Craft House at Tehran University of Art | 2021
Supervisor: Dr. Alireza Mostaghni
Team Project | Contribution: Form Finding, Execution

Freeform Space Structure Workshops are held for the purpose of researching and practicing the two areas of free-form structures and computer-integrated construction and design methods. Each of these training programs deals with a specific way of constructing free-form structures. This workshop tried to design and construct a structure by examining surface active and bending active structures. At first, students have created ways to develop the form by studying the materials and understanding their behavior. Then, they developed their designs using digital fabrication tools and produced them on a real scale. All parts of these two structures are made of 6mm thick plywood and cut using digital fabric cutting machines. Then, by applying force and using the elastic properties of the materials, they are bent to their final shapes.

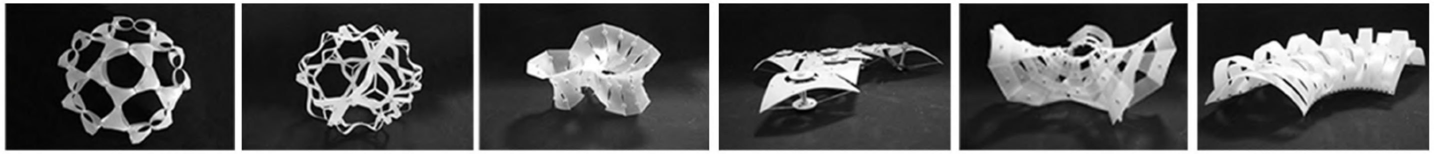
Process

In the zero phase of the workshop, different methods of fabrication of these types of structures and different types of materials were tested for their elastic deformation and ability to be cut with digital fabric-cutting machines. As a result of this process, "cardboard and high Impact polystyrene" was selected as the desired prototyping material, and plywood with different thicknesses for use in the final work.

The first phase of the workshop consisted of two parts: training and experience. In the training section, students were introduced to the Grasshopper software as a digital design tool and digital fabrication tool to realize their potential in design. Students were then asked in the experience section to try to bend a sheet to form a module or part of an active bend structure. In this section, the students succeeded in designing seven different prototypes. In the next step, three different systems were selected as the main options, and students studied and developed them in three groups.

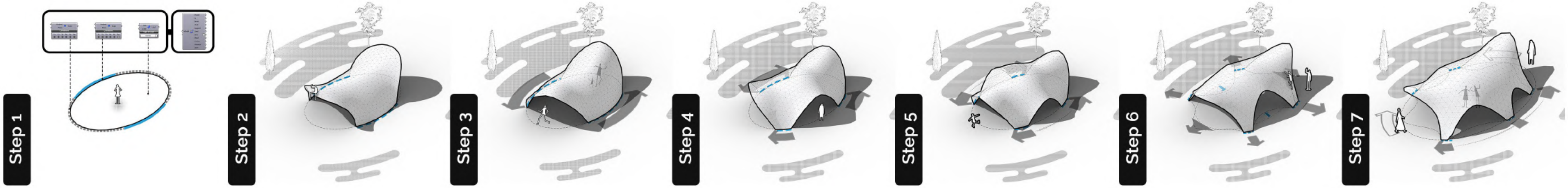
The output of this part was a prototype of a fifth of the whole or part of the final work. This is to ensure that each group experiences the process of producing a prototype of the pavilion using digital design tools and in the same detail as the final work.

In the last step, a design alternative was selected by students' vote and was made on a real scale. The production of construction drawings was made entirely by code in the Grasshopper software, and all parts were produced by CNC milling and laser cutting.

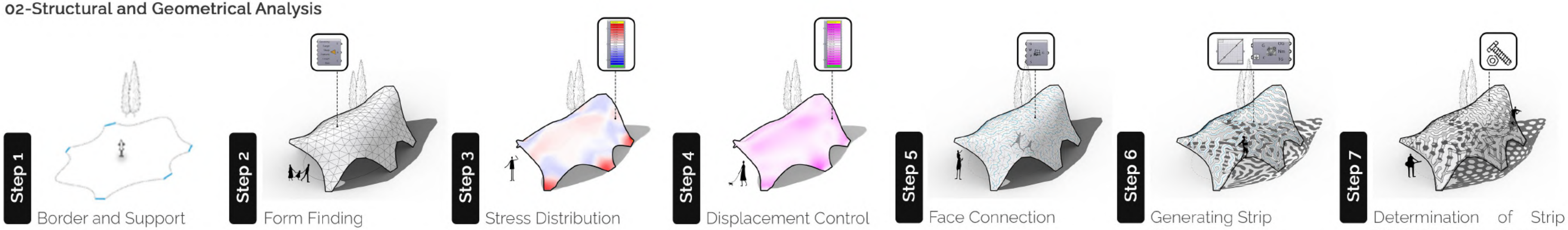


Digital Design Process

01-Form Making



02-Structural and Geometrical Analysis



Border and Support

Form Finding

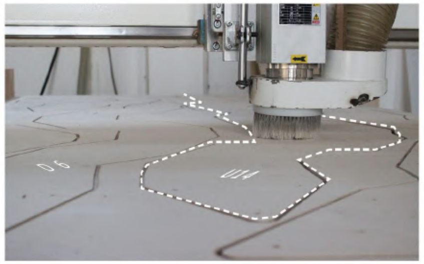
Stress Distribution

Displacement Control

Face Connection

Generating Strip Elements

Determination of Strip Width Base on Structural Analysis



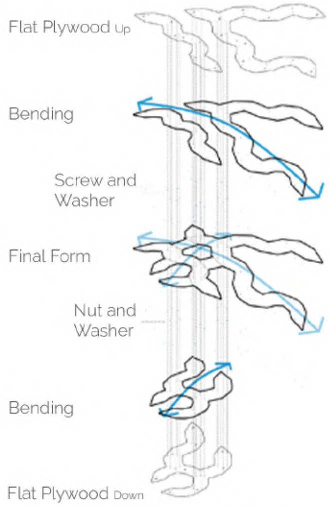
Sheet No.4

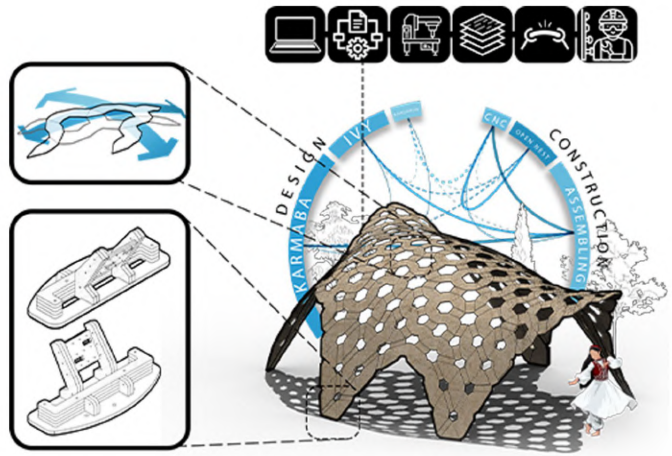
Sheet No.5

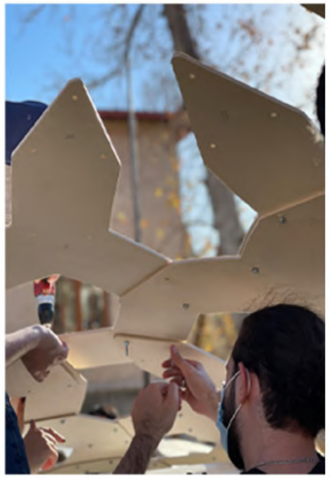
Sheet No.6

Fabrication Process

The CNC milling machine cut the elements of this project from the 6mm thick plywood sheets. Although the final 3D model contains bent elements, these elements were fabricated as flat pieces, creating the final shape regarding the accurate joint position. In other words, the position of the joints was calculated based on the final form of the structure. Therefore, it is impossible to bolt them unless the elements are in their final shape and position.







Final Product

The product of this workshop is the pavilion that is placed in the courtyard of the Bagh-Melli campus. This pavilion is made of strip-like elements that are connected in two perpendicular directions and provide a three-dimensional bent in space, and are stabilized in the same way. The final shape is the result of putting these pieces together. Initially, this form was digitally modeled on Grasshopper software, and then the unroll of each piece was plotted. Due to the five-meter length of the pieces, each piece was split into smaller pieces so that they could be cut from standard plywood sheets 122 x 244 cm. Also, some members were produced as a stitching plate for these smaller parts to produce a U-shaped form and to bind the U-shapes together.

07

Printed Structure

Freeform Space Structure (FSS #6)

Workshop | Held by Digital Craft House at Tehran University of Art | 2022
Supervisor: Dr. Alireza Mostaghni
Team Project | Contribution: Form Finding, Execution

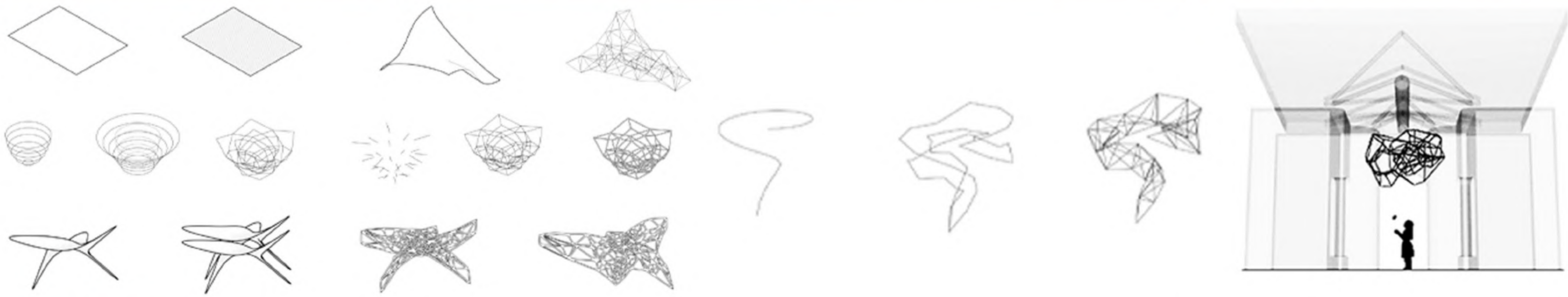
This workshop focused on two topics, 3D printing technology and computational design and fabrication of the freeform standardless spatial structure. To learn and practice 3D printing, the workshop started with a simple object fabrication. Then, students try to design and fabricate the different types of joints. Next, we developed a BIM tool to convert a wire mesh to a LOD400 BIM model and fabrication data. Finally, researchers fabricate the elements and assemble the structure.

Prototyping

In this section, attendees designed and developed a system or a joint that can connect the structural elements to each other, according to what they learned in the first chapter. Then, they fabricated these prototypes, and students selected the best alternative considering criteria like aesthetics and strength.

Form Design

Researchers tried to generate different wire meshes as the form for the final product of the workshop, which was the real scale structure. Various forms were developed and designed in relation to the design systems. Finally, the best form, which was a kind of cellular 3D Voronoi selected by the researchers.

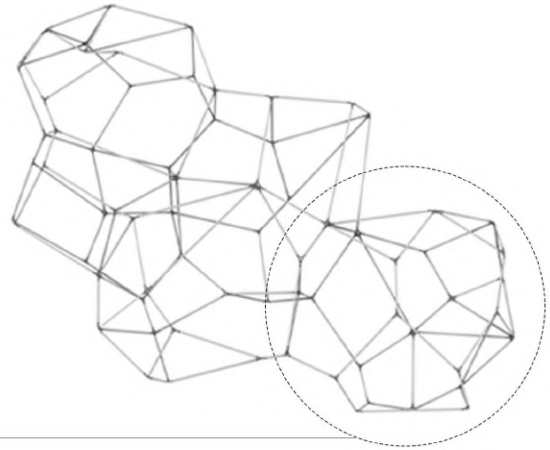


Final Product

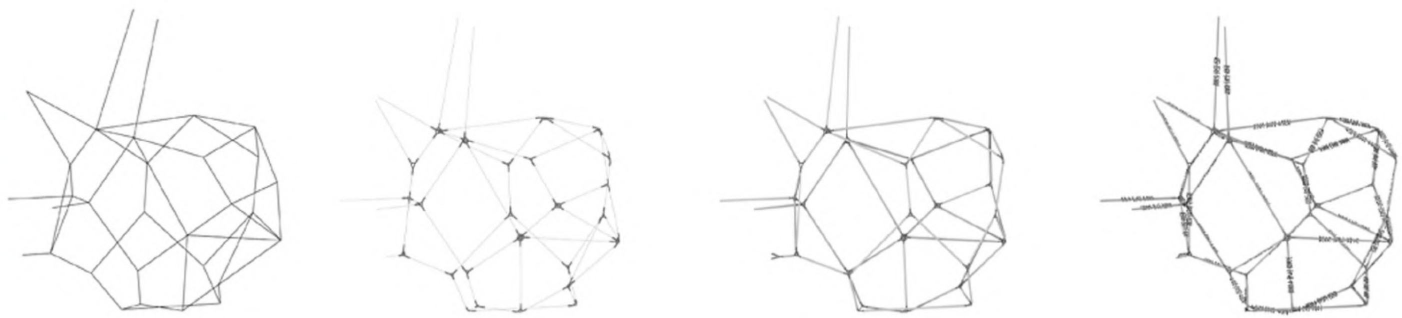
We had a wire mesh as a form and a system, and we wanted to convert these data to a real structure. First, we developed a computational BIM generator tool that converts a wire mesh to a detailed BIM model containing fabrication data. Then, structural elements and joints are fabricated using digital fabrication tools, and the components are assembled to create the final structure.

A. BIM Modeling

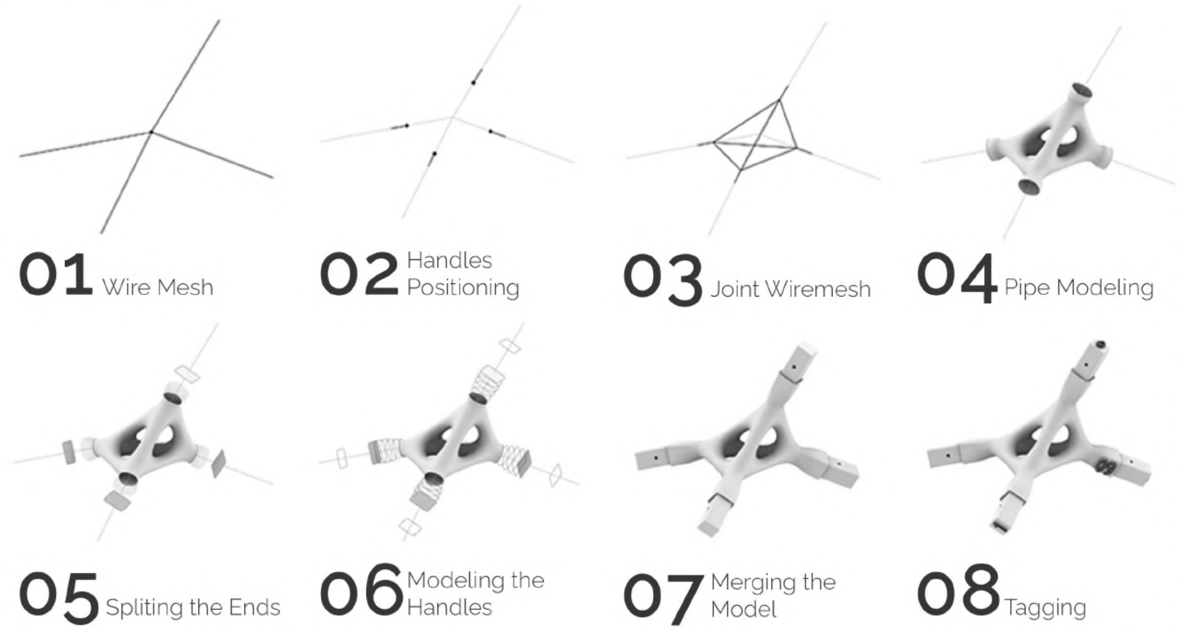
A model containing all of the elements, joints, bolts, and accurate information was needed to fabricate the structure, which is hard to create when the design is freeform. Thus, we developed a computational tool that uses the system logic to convert the wire mesh to the final model immediately and automatically.

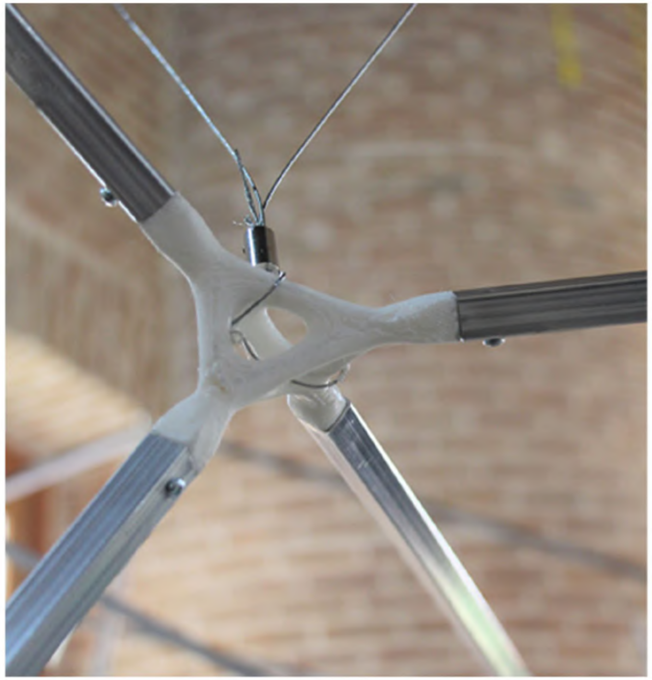


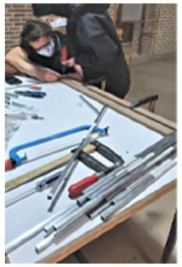
01 Wire Mesh — **02** Joint Modeling — **03** Element Modeling — **04** Tagging



B. Joint Modeling







C. Fabrication

The joints, the most controversial part of this structure, were fabricated using several 3D printers in 5 days. Whereas the only variable property of the elements was their length, elements were cut in different lengths from long metal profiles.

D. Assembly

In the end, the structure was assembled in a day by the students and hung from the ceiling of the hall of the University of art.